

Employee Attrition Prediction – Executive Summary

Project Summary

Employee turnover is a major challenge for organizations because replacing experienced employees requires significant time, effort, and cost. The objective of this project was to identify employees who may be at a higher risk of leaving the company so that HR can take preventive action before valuable talent is lost.

The analysis found that employee attrition is influenced by a combination of factors rather than salary alone. Employees with shorter tenure, lower job satisfaction, poor work-life balance, frequent overtime, and lower monthly income were more likely to leave the organization. Certain job roles and departments also experienced higher attrition than others, indicating that retention efforts should be targeted rather than applied uniformly across the company.

Using historical employee data, a predictive system was developed to estimate the likelihood of an employee leaving. The results demonstrate that employee records can be used to identify patterns associated with attrition, allowing HR teams to focus on employees who may benefit from additional support, career development, or engagement initiatives.

Based on these findings, the company should prioritize improving employee engagement during the first few years of employment, regularly monitor job satisfaction and work-life balance, review overtime practices, and create structured career growth opportunities for employees in high-risk roles. Conducting periodic one-on-one retention discussions with employees identified as higher risk may further reduce voluntary resignations.

It is important to note that this system should be used as a decision-support tool rather than a replacement for managerial judgment. Employee decisions are influenced by personal circumstances, organizational changes, and external opportunities that may not be reflected in historical data. Therefore, the predictions should be combined with regular employee feedback, performance discussions, and HR expertise to make informed retention decisions.

Overall, this project demonstrates how data-driven insights can help the HR department proactively identify potential attrition risks, reduce recruitment and training costs, improve employee retention, and support better workforce planning.